

ORIGINAL RESEARCH

GENETICS, OMICS, AND TISSUE REGENERATION

Unbiased Discovery of Genetic Determinants of Resilience to CAD

Insights From PROMISE and CATHGEN



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ABSTRACT

BACKGROUND Genetic determinants of resilience remain poorly defined beyond family studies.

OBJECTIVES The purpose of this study was to perform an unbiased study of individuals with discordance between clinical/genetic risk and atherosclerotic burden to discover novel genetic pathways underpinning atherosclerosis.

METHODS We used 2 genotyped cohorts with well-defined coronary anatomy: PROMISE (Prospective Multicenter Imaging Study for Evaluation of Chest Pain) (discovery cohort: coronary computed tomography angiography) and CATHGEN (CATHeterization GENetics) (validation cohort: invasive angiography). Resilience was defined as high clinical and polygenic risk of coronary artery disease (CAD), yet without coronary plaque. Resilient individuals were compared to patients with obstructive CAD (oCAD) (stenosis $\geq 70\%$) using genome-wide association analyses at variant, gene, and pathway levels.

RESULTS In PROMISE ($n = 605$), 46 (8%) were resilient and 88 (15%) had oCAD. In CATHGEN ($n = 3,236$), 127 (4%) were resilient and 1,852 (57%) had oCAD. Clinical risk factors and polygenic risk scores were similar between resilient and oCAD patients in both cohorts. Variant- and gene-level analyses did not yield genome-wide significant signals. Pathway-level analyses identified 4 resilience-associated pathways in PROMISE that replicated in CATHGEN: adipocytokine signaling, fatty acid metabolism, fatty acid degradation, and vascular smooth muscle contraction.

CONCLUSIONS Resilience to CAD—defined as the absence of coronary atherosclerosis despite high clinical and polygenic risk—is present in both lower- (PROMISE) and higher-risk (CATHGEN) cohorts and is linked to protective variants in metabolic and vascular pathways. This unbiased, proof-of-concept approach reveals biologically plausible targets for replication and mechanistic studies in larger imaging-based genetics cohorts. (JACC Adv. 2026;5:102665) © 2026 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**ABBREVIATIONS
AND ACRONYMS****CAD** = coronary artery disease**CCTA** = coronary computed tomography angiography**FH** = familial hypercholesterolemia**GWAS** = genome-wide association studies**LDL** = low-density lipoprotein**Lp(a)** = lipoprotein a**oCAD** = obstructive coronary artery disease**PRS** = polygenic risk score

Coronary artery disease (CAD) and its complications (myocardial infarction, heart failure, and sudden death) are a leading cause of morbidity and mortality worldwide.¹ Despite risk factor control, many patients still develop CAD. Conversely, other patients—despite a high burden of genetic and clinical risk factors—are resilient to atherosclerosis and do not develop CAD. These resilient patients may have unique molecular characteristics as an “extreme” phenotype conferring resilience to CAD. Studying extreme phenotypes is a well-established method to uncover molecular pathways which can form the basis of

novel and transformative therapies to reduce cardiovascular disease risk.^{2–4} However, no study has evaluated genetic resilience to CAD in well-characterized cohorts at both lower and higher risk of CAD using core lab-confirmed measures of CAD severity.

Genome-wide association studies (GWAS) demonstrate that the genetic basis for CAD is predominantly due to multiple common genetic variants, each with small effect sizes.⁵ These common variants can be represented by a polygenic risk score (PRS), a quantitative factor that captures the aggregate genetic influence of multiple common genetic variants. A PRS can capture genetic risk for CAD independent of traditional risk factors and may have clinical utility.⁶ In addition to factors that *increase* risk for CAD, there are recent data on factors associated with a *lower* CAD risk in patients with risk factors but without CAD—a paradigm known as “resilience.” One example of CAD resilience which has had significant clinical impact are patients with genetic variants in *PCSK9* that confer lower levels of low-density lipoprotein (LDL) and lifelong protection from CAD.³ Additional insights into resilience come from carriers of pathogenic variants causing familial hypercholesterolemia (FH) but are without CAD—in whom increased high-density lipoprotein (levels and gene expression pathways) and lower lipoprotein a [[Lp(a)] have emerged as mitigating the risk conferred by FH variants.⁷ Beyond patients with FH variants, a genome-wide association study of septuagenarians without prevalent or incident CAD identified regulatory variants in *CETP*—a well-known regulator of high-density lipoprotein, LDL, and Lp(a)—as cardioprotective.⁸ There are also nongenetic determinants of cardiovascular resilience, including environmental and lifestyle factors such as a healthy diet and favorable social determinants of health.^{9,10}

In this analysis, our goal was to identify novel molecular pathways of resilience to CAD. We hypothesized that genetic loci and molecular pathways protective against CAD are active in patients with a high burden of CAD risk factors yet without coronary atherosclerosis by coronary computed tomography angiography (CCTA) or invasive angiography. To test our hypothesis, we leveraged established biorepositories including genetic data and core lab-confirmed CCTA results from participants in the PROMISE (Prospective Multicenter Imaging Study for Evaluation of Chest Pain; [NCT01174550](https://clinicaltrials.gov/ct2/show/study?term=NCT01174550)) trial and validated our findings among participants in the CATHGEN (CATHeterization GENetics) cohort.^{11,12}

METHODS

OVERVIEW. The Duke Institutional Review Board has approved the analysis and distribution of these data for the purpose of publication. PROMISE clinical data sets are available at <https://biolincc.nhlbi.nih.gov/studies/promise>. There are no commercial use data restrictions, and no data restrictions based on area of research. Genetic data from PROMISE are available upon reasonable request. Molecular data from CATHGEN are available through the NCBI dbGaP database (accession phs000703.v1.p1), and clinical data from CATHGEN are available upon reasonable request.

We used the PROMISE biorepository cohort to conduct genetic association analyses to identify putative genes/pathways associated with resilience, as CCTA is most reliable for ruling out coronary artery plaque. The CATHGEN cohort was then used for independent validation of genes/pathways identified in PROMISE ([Supplemental Figure 1](#)). As an overview, we first defined resilience in PROMISE as those participants with high clinical and/or genetic probability for CAD but with no coronary plaque on CCTA. Next, we compared “resilient” PROMISE biorepository participants to all PROMISE biorepository participants with genetic data and obstructive CAD (oCAD) on CCTA using variant-, gene-, and pathway-level analyses. Finally, we performed independent validation in the CATHGEN biorepository. To do this, we used a comparable approach to identify resilient participants in CATHGEN and conducted gene- and pathway-level analyses to validate the genetic signature of resilience identified in PROMISE. In an exploratory analysis to gain additional insight into resilience, we also defined a phenotype of susceptibility, defined as participants with oCAD despite low

clinical and genetic risk. We substituted this “extreme” comparator for oCAD as an additional comparison against the resilient phenotype in PROMISE.

ETHICAL CONSIDERATIONS. Both the PROMISE and CATHGEN studies were conducted in accordance with the Declaration of Helsinki and were approved by the necessary ethics committees. The PROMISE study received approval from the Institutional Review Boards (IRBs) at Duke, Partners Healthcare, and Tufts, along with local IRBs at each of the 193 participating sites in North America. The CATHGEN study was approved by the Duke University IRB. All participants provided informed consent.

PARTICIPANT COHORTS. PROMISE cohort. Design and primary results of the PROMISE trial have been published.^{12,13} PROMISE randomized stable outpatients with suspected CAD to either an initial anatomic testing strategy using CCTA vs functional testing (imaging or nonimaging exercise or pharmacologic stress testing). In this nested cohort study from the PROMISE trial, we included all participants randomized to the CCTA arm, who received the initial diagnostic test as randomized and had available genotyping data (Supplemental Figure 1).¹³ As previously published, of the 10,003 PROMISE participants, 4,070 participated in the PROMISE biorepository.¹⁴ We started with the 692 patients with available genetic data from the biorepository; of these, 620 had analyzable CCTAs by the core lab.¹⁴ We excluded $n = 44$ with nonobstructive CAD <70% stenosis, for a final analysis PROMISE cohort of 544 (Supplemental Figure 2). We have previously found that biorepository participants were similar to overall PROMISE trial participants.¹⁴

Determination of obstructive CAD and no CAD in PROMISE. Central core lab-based interpretations of study CCTAs were used, and oCAD was defined as $\geq 70\%$ stenosis in any major epicardial coronary artery or $\geq 50\%$ left main stenosis, as previously described.¹⁵ Patients without CAD were defined as having no core lab-identified coronary plaque in any coronary artery. Overall, there were 80 patients (15%) with oCAD and 464 patients (85%) with no CAD available for this analysis (Supplemental Figure 2).

Definition of resilience in PROMISE. Conceptually, our intent was to identify patients at high clinical and genetic risk without coronary plaque on CCTA, our functional definition of resilience (Central Illustration). We used a multistep process to achieve this goal.

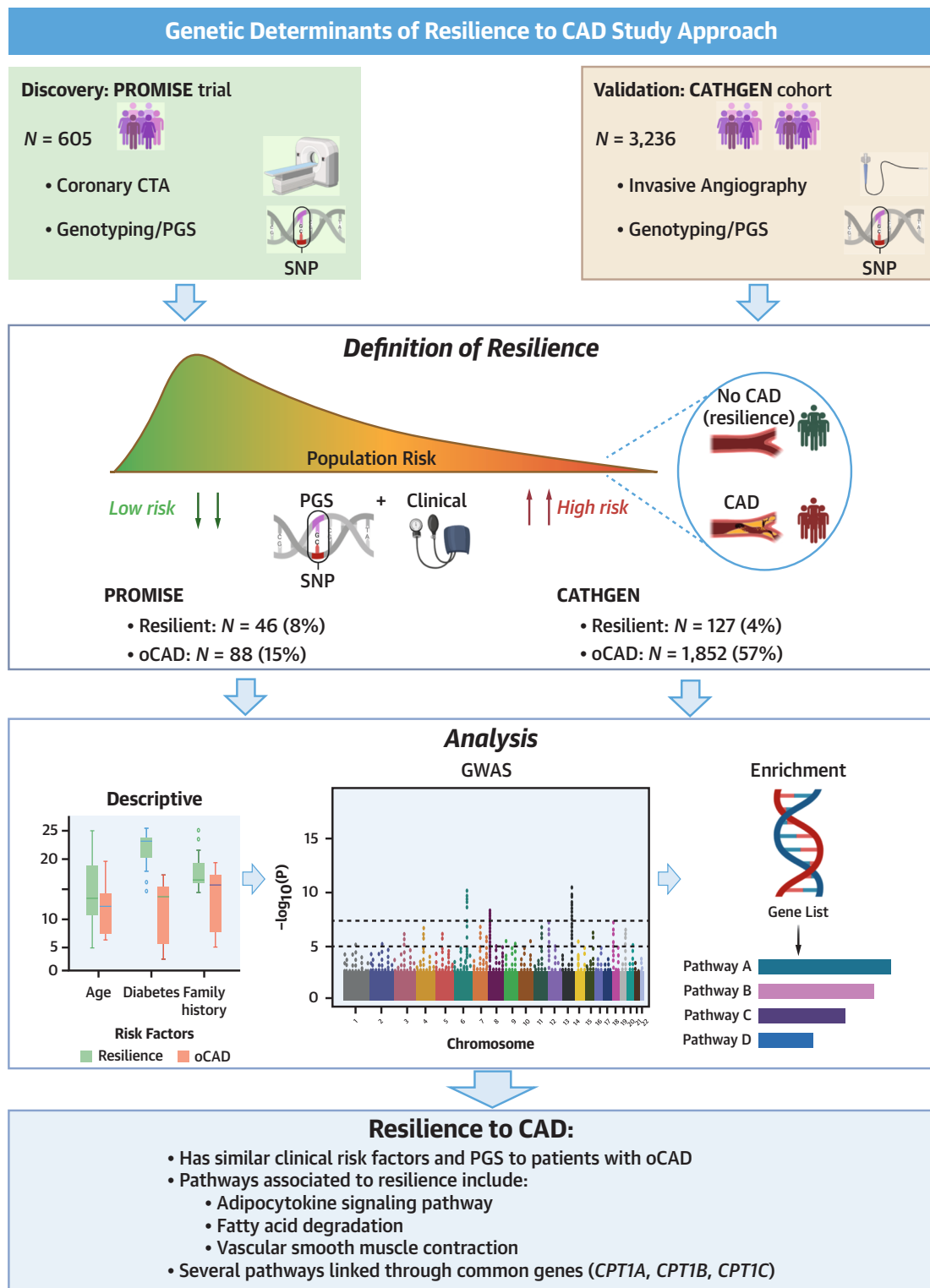
In step 1, we generated a multivariable model to predict oCAD that considered a broad range of

possible clinical risk factors: age, sex, diabetes, hypertension, dyslipidemia, family history of CAD, smoking, the pooled cohort equations atherosclerotic cardiovascular disease¹⁶ and Framingham Risk scores,¹⁷ high-sensitivity C-reactive protein, LDL particle number (adjusted for statin use) and statins,¹⁸ the 2019 European Society Cardiology pretest probability score, and the PRS. This model included participants with oCAD ($n = 80$) or no CAD ($n = 464$), a PRS for oCAD (PRS),¹⁹ and complete clinical data. Variable selection was performed using a mixed approach (forward - backward) by adding or excluding the most/least significant factor while forcing in the PRS. Ten-fold cross-validation was performed to reduce risk of overfitting and metrics including R^2 and Akaike Information Criterion were used to achieve the model that best predicted oCAD. The final model included the PRS along with age, sex, family history of CAD, Framingham risk score, high-sensitivity C-reactive protein, adjusted LDL, and statin use (area under receiver-operating characteristics curve = 0.85; 95% CI: 0.80-0.90, $R^2 = 0.36$) (Supplemental Table 1). We interpreted the predictive probabilities for oCAD generated by the model to be the combined effects of clinical and genetic risk factors for oCAD.

In step 2, we used these probabilities as a “score” to select patients with no CAD based on their predicted risk for oCAD. We generated separate histograms of the predicted probability for patients with and without oCAD (Supplemental Figure 4). We visually inspected the histograms to identify patients from each group with a similar predicted probability of oCAD and of adequate size for GWAS analysis. Out of the total 464 patients with no CAD, we selected 46 ($\approx$ top 10%) with the highest predicted probability of oCAD to define resilience.

Definition of susceptibility in PROMISE. Similar to the approach to define resilience, we used the predictive probabilities of the clinical and genetic risk model to define susceptibility (ie, presence of oCAD despite low predicted risk). Out of the total 80 patients with oCAD, we selected $n = 32$ ($\approx$ bottom 40%) with the lowest predicted probability of oCAD to define susceptibility.

CATHGEN COHORT. The structure of the Duke Catheterization Genetics (CATHGEN) cohort has been previously described.^{11,20} Patients presenting for coronary angiography at Duke University Medical Center were enrolled into a biorepository. This analysis includes CATHGEN biorepository participants with available angiographic and genotyping data (Supplemental Figure 1C, Supplemental Figure 3). Of the 9,781 CATHGEN participants, 3,235

CENTRAL ILLUSTRATION Definition of Resilience

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"Resilient" participants were defined as those with high clinical and/or genetic risk for CAD, but with no coronary plaque on coronary imaging (CTA or angiography). CTA = computed tomography angiography; GWAS = genome-wide association studies; other abbreviations as in [Figure 1](#).

had genome-wide genotyping and angiographic data available.

Determination of obstructive CAD and no CAD in CATHGEN. In CATHGEN, oCAD was defined as $\geq 70\%$ of any major epicardial coronary artery or $\geq 50\%$ left main stenosis from clinically interpreted angiograms. Patients without any CAD were defined as having $<10\%$ stenosis in all coronary arteries (the lowest category on the data collection form) and no prior history of CAD. There were 1,852 CATHGEN patients with oCAD and 637 patients without any CAD (Supplemental Figure 3).

Definition of resilience in CATHGEN. For validation, we repeated the processes used in PROMISE to define resilience in CATHGEN. The initial covariates were age, sex, race, body mass index, diabetes, family history of CAD, smoking, dyslipidemia, hypertension, and PRS (described below). Following the same variable selection process as in PROMISE, the final model covariates included PRS, age, sex, blood pressure, diabetes, family history of CAD, smoking, dyslipidemia, body mass index, and race (area under receiver-operating characteristic curve = 0.84; 95% CI: 0.83–0.86, $R^2 = 0.32$) (Supplemental Table 2). Similar to PROMISE, in CATHGEN, we visually inspected the histograms to identify patients from each group with a similar predicted probability of oCAD and of adequate size for GWAS analysis (Supplemental Figure 5). Out of the total of $N = 638$ without CAD in CATHGEN, we selected $N = 127$ ($\approx$ top 20%) with the highest predicted probabilities of oCAD to define resilience.

COMPUTING POLYGENIC RISK SCORE. The PRS was computed in the PROMISE cohort as previously described¹⁴ using imputed genotype data from sequenced samples. We applied validated PRS weights¹⁹ and calculated PRS in PLINK v1.90b10 (–score) by multiplying the risk-allele dosage at each variant by its corresponding weight and summing across all variants ($N = 5,563,114$). Genotypes were imputed via the Michigan Imputation Server with 1,000 Genomes Phase 3 v5 reference panel.²¹ Following prior methodology,¹⁴ we adjusted for population structure using the first 5 genetic principal components. Using available genetic data, the same procedures were used to calculate a PRS in CATHGEN.²⁰

GENERAL STATISTICAL METHODS. Comparisons of clinical characteristics and PRS between resilient patients and those with oCAD were compared using *t*-tests or chi-square, as appropriate. A Wilcoxon test was used to compare differences between predicted probabilities of oCAD across 3 groups: 1) resilient; 2)

oCAD; and 3) no CAD, nonresilient in PROMISE and CATHGEN. All analyses were conducted using R statistical software²² unless stated otherwise.

GENETIC ASSOCIATION AND PATHWAY ANALYSIS

IN PROMISE. We first sought to identify single-nucleotide polymorphisms (SNPs) associated with resilience in PROMISE with imputed genotype data using Plink and logistic regression adjusted for age, sex, race, and the first 5 genetic principal components.²³ In this analysis, we compared resilient patients to all participants with oCAD, as defined above. Next, we sought to identify genes associated with resilience in PROMISE using GCTA fastBAT²⁴ to aggregate test statistics for individual variants at the gene level. We used a nominal *P* value threshold of 0.05 for significant genes in this model. Next, the 1,272 genes meeting this *P* value threshold were used for enrichment over-representation analysis with R-package ClusterProfiler²⁵ and KEGG database²⁶ to identify biologically relevant pathways associated with these genes using a false discovery rate adjusted *P* value <0.05 . Similar pathway analyses were also completed for the exploratory analysis comparing the resilient and susceptible phenotypes.

VALIDATION WITH CATHGEN COHORT. In order to validate the genes and pathways identified in PROMISE, we repeated the SNP, gene, and pathway-level analyses in CATHGEN. For validation, we evaluated the performance of genes and pathways at a nominal *P* value <0.05 because the pathways tested in CATHGEN were limited to those identified in PROMISE at a false discovery rate adjusted *P* value <0.05 , reducing the multiple-testing burden.

RESULTS

Among the $N = 605$ participants from PROMISE with available genetic and CCTA data, we identified $n = 46$ resilient participants and $n = 80$ participants with oCAD (Supplemental Figure 2). Compared with PROMISE participants with oCAD, resilient patients had similar risk factors for CAD and a similar PRS (unadjusted *P* values >0.10), (Table 1). Similarly, among the $N = 3,235$ in CATHGEN with available genetic and angiographic data, we identified $n = 127$ resilient patients and $n = 637$ patients with oCAD (Supplemental Figure 3). With the exception of family history of CAD, the distribution of clinical risk factors and PRS was similar between resilient patients and those with oCAD in CATHGEN (Table 1). Notably, in both PROMISE and CATHGEN, the predicted clinical + PRS probability of oCAD was similar for resilient patients compared to those with oCAD (Figure 1), demonstrating the robustness of our

TABLE 1 Baseline Characteristics of Resilient and oCAD Patients in PROMISE

	PROMISE			CATHGEN		
	Resilient (n = 46)	oCAD (n = 80)	P Value	Resilient (n = 127)	oCAD (n = 1,852)	P Value
Age, mean (SD), y	61.44 (8.34)	62 (7.95)	0.71	60.77 (10.44)	62.07 (11.54)	0.22
Female, n (%)	9 (19.57%)	22 (25%)	0.48	45 (35.43%)	681 (36.77%)	0.76
Race/ethnicity			0.21			0.79
White (Caucasian), n (%)	38 (82.61%)	81 (92.05%)		90 (70.87%)	1,379 (74.46%)	
Black, n (%)	5 (10.87%)	4 (4.55%)		26 (20.47%)	322 (17.39%)	
Other race, n (%)	3 (6.52%)	3 (3.41%)		11 (8.66%)	151 (8.15%)	
Cardiovascular risk factors						
BMI, mean (SD)	29.26 (5.83)	30.56 (5.02)	0.19	30.74 (7.51)	29.92 (6.33)	0.17
Hypertension, n (%)	36 (78.26%)	60 (68.18%)	0.22	88 (69.29%)	1,339 (72.3%)	0.47
Diabetes, n (%)	13 (28.26%)	27 (30.68%)	0.77	47 (37.01%)	601 (32.45%)	0.29
Dyslipidemia, n (%)	37 (80.43%)	63 (71.59%)	0.27	87 (68.5%)	1,278 (69.01%)	0.91
Family history of premature CAD, n (%)	16 (34.78%)	31 (35.23%)	0.96	32 (25.2%)	773 (41.74%)	<0.001
Current or former tobacco use, n (%)	25 (54.35%)	56 (63.64%)	0.30	64 (50.39%)	1,006 (54.32%)	0.39
Statin use, n (%)	30 (65.22%)	47 (53.41%)	0.19	N/A	N/A	
Aspirin use, n (%)	46 (100%)	84 (95.45%)	0.28	N/A	N/A	
ASCVD risk score (mean, SD)	19.7 (8.6)	21.28 (15.65)	0.52	N/A	N/A	
ESC PTP score (mean, SD)	21.33 (9.49)	21.26 (9.9)	0.97	N/A	N/A	
LDLc (mean, SD) mg/dL	122.02 (39.18)	124.05 (34.89)	0.76	103.49 (32.66)	99.36 (41.19)	0.41
Adjusted LDLc (mean, SD) mg/dL	131.62 (40.67)	131.91 (32.78)	0.97	N/A	N/A	
Hs C-reactive protein (mean, SD) mg/L	6.01 (10.77)	6.95 (18.21)	0.75	N/A	N/A	
PRS (SD)	0.39 (1.02)	0.52 (1.09)	0.48	0.19 (0.86)	0.17 (1.01)	0.89

Statistical analyses were conducted using linear, logistic regression or ANOVA, as appropriate based on variable type (continuous or categorical).
ANOVA = Analysis Of Variance; ASCVD = atherosclerotic cardiovascular disease; BMI = body mass index; CATHGEN = CATHeterization GENetics; ESC PTP = European Society of Cardiology pretest probability; LDLc = low-density lipoprotein cholesterol; oCAD = obstructive coronary artery disease; PRS = polygenic risk score; PROMISE = Prospective Multicenter Imaging Study for Evaluation of Chest Pain.

functional definition of resilience to oCAD across 2 cohorts and suggesting the presence of unmeasured protective factor(s).

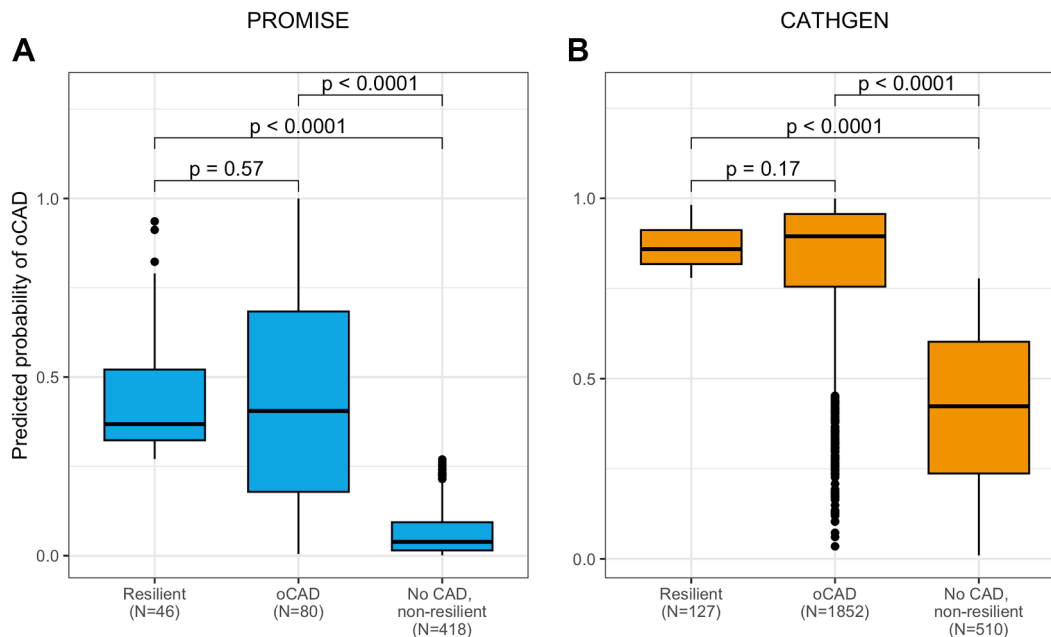
To identify genetic factors of resilience in PROMISE, we conducted a GWAS of the 46 resilient patients compared with the 80 patients with oCAD. Single-variant and gene-level GWAS did not identify variants or genes exceeding genome-wide significance (Supplemental Figures 6 to 9). We next performed pathway analysis of the genes that emerged from gene-level GWAS with an unadjusted $P < 0.05$ ($n = 1,272$). We identified 7 pathways (enrichment $P_{adj.} < 0.05$) associated with resilience (Table 2). Approximately 15% (13% to 17%) of genes in each pathway had nominally significant gene-level associations with resilience (Table 2). Several of these pathways were linked through common genes, for example, *CPT1A*, *CPT1B*, and *CPT1C* are common to adipocytokine signaling, glucagon signaling, alcoholic liver disease, and fatty acid metabolism/degradation pathways.

Next, we validated the pathways associated with resilience in PROMISE by repeating these analyses in CATHGEN. We performed the same gene-level GWAS

and pathway enrichment in CATHGEN as was performed in PROMISE. Among the 7 pathways associated with resilience in PROMISE, 4 were also significant in CATHGEN: adipocytokine signaling, fatty acid metabolism, fatty acid degradation, and vascular smooth muscle contraction (nominal enrichment $P < 0.05$) (Table 2). Despite validation at the pathway level, individual genes in each pathway demonstrated only partial overlap between the cohorts (Table 2). Thus, despite the observation that the genes within each pathway differed across cohorts, the pathways associated with resilience were consistent in both PROMISE and CATHGEN.

In an effort to identify additional pathways associated with resilience, exploratory analyses were conducted in PROMISE comparing resilient patients with those defined as susceptible. Using this more extreme comparison using susceptibility (oCAD despite low predicted clinical and genetic risk), 3 additional pathways (Pertussis, Retinol Metabolism, Olfactory transduction) (Supplemental Table 3) were identified at an adjusted P value < 0.05 , in addition to the original 7 pathways identified in the resilient vs oCAD comparison.

FIGURE 1 Predicted Probabilities by CAD Phenotype in PROMISE and CATHGEN



A and B show box-and-whisker plots for resilient, obstructive CAD (oCAD), and no CAD groups (x-axis). The y-axis is the predicted probability. Boxes show the IQR with the median. Whiskers extend to the most extreme values within $1.5 \times \text{IQR}$ of Q1 and Q3; points beyond are plotted as outliers. *P* values are from the Wilcoxon test. Only participants with complete clinical data needed to compute probabilities are included. CAD = coronary artery disease; CATHGEN = CATHeterization GENetics; oCAD = obstructive coronary artery disease; PROMISE = Prospective Multicenter Imaging Study for Evaluation of Chest Pain.

DISCUSSION

While much is known about the molecular underpinnings of atherosclerosis and CAD, the absence of CAD in those at apparent high risk (or the presence of disease in those at low risk) has long intrigued clinicians and researchers. Investigation of such “outlier” populations has advanced understanding of vascular biology and has identified druggable pathways supporting the development of novel preventive therapies with proven clinical effectiveness.^{4,27} Beyond CAD, the availability of large cohorts of patients with linked electronic medical record and genetic data has identified those who “should” have monogenic diseases, yet lack clinical manifestations.²⁸ This phenomenon has been labeled as “incomplete penetrance” without further systematic investigation. Notably, individuals with “incomplete penetrance” of diseases with a strong genetic component may inform novel paradigms for identification of disease modifying or protective factors. Furthermore, longitudinal cohorts of such patients could support

robust phenotyping and advance understanding of the molecular underpinnings of resilience.

In this current work, we leveraged 2 large cohorts at different points along the spectrum of clinical risk. This analysis identified and validated 4 pathways associated with high clinical and genetic risk, but with little to no coronary plaque: adipocytokine signaling, fatty acid metabolism/degradation, and vascular smooth muscle contraction. These findings confirm our hypothesis that patients with a high probability of CAD based on clinical and genetic risk factors but without CAD are enriched for protective genetic variants and molecular pathways when compared to similar-risk patients with oCAD.

Our findings advance prior knowledge of the metabolic underpinnings of CAD and show that patients resilient to traditional clinical risk factors and polygenic risk for CAD are characterized by protective alterations in common metabolic and vascular pathways. The role of adipocytes and fatty acids in the development of CAD has been described previously.^{29,30} We build on this finding by identifying

TABLE 2 KEGG Canonical Pathways Associated With Resilience in PROMISE and CATHGEN

KEGG Canonical Pathway	Gene Ratio (Fold Enrichment ^a)		P Value		Gene Symbol	
	PROMISE	CATHGEN	PROMISE ^a	CATHGEN ^b	PROMISE	CATHGEN
Adipocytokine signaling pathway	0.03 (3.23)	0.02 (1.71)	<0.01	0.02	<i>CPT1C</i> , <i>SOCS3</i> , <i>NFKB1B</i> , <i>CPT1A</i> , <i>NFKB1E</i> , <i>CPT1B</i> ^c , <i>PRKAA2</i> , <i>PRKAG3</i>	<i>RXRA</i> , <i>ADIPOR2</i> , <i>MAPK10</i> , <i>CPT1B</i> , <i>NPY</i> , <i>ACSL3</i>
Alcoholic liver disease	0.05 (2.47)	0.03 (1.84)	<0.01	0.09	<i>CPT1C</i> , <i>IRF3</i> , <i>CD14</i> , <i>CPT1A</i> , <i>CXCL8</i> , <i>CIQC</i> , <i>CIQB</i> , <i>MAP3K14</i> , <i>CIQA</i> , <i>RIPK1</i> , <i>FOXO3</i> , <i>CPT1B</i> , <i>PRKAA2</i> , <i>PRKAG3</i>	<i>ADIPOR2</i> , <i>MAPK10</i> , <i>CPT1B</i> , <i>ALDH1B1</i> , <i>ALDH9A1</i> , <i>LEF1</i> , <i>C4A</i> , <i>C4B</i> , <i>C2</i> , <i>IL1B</i> , <i>MAPK13</i>
Glucagon signaling pathway	0.04 (2.72)	0.02 (1.86)	0.001	0.22	<i>CPT1C</i> , <i>PRMT1</i> , <i>PGAM1</i> , <i>CPT1A</i> , <i>SLC2A2</i> , <i>CPT1B</i> , <i>PFKM</i> , <i>PRKAA2</i> , <i>PRKAG3</i> , <i>PPP3R2</i> , <i>SIK1</i> , <i>PDE3B</i>	<i>CAMK2D</i> , <i>CALM1</i> , <i>CPT1B</i> , <i>ITPR3</i> , <i>PPP3R1</i>
Fatty acid metabolism	0.03 (3.04)	0.02 (1.94)	0.008	0.03	<i>CPT1C</i> , <i>CBR4</i> , <i>ELOVL2</i> , <i>CPT1A</i> , <i>EHHADH</i> , <i>CPT1B</i>	<i>CPT1B</i> , <i>SCP2</i> , <i>ACSL3</i> , <i>CBR4</i>
Fatty acid degradation	0.02 (3.11)	0.01 (1.40)	0.026	0.04	<i>CPT1C</i> , <i>CPT1A</i> , <i>EHHADH</i> , <i>CPT1B</i>	<i>CPT1B</i> , <i>ALDH1B1</i> , <i>ALDH9A1</i> , <i>ACSL3</i> , <i>ACADS</i>
Vascular smooth muscle contraction	0.04 (2.15)	0.03 (1.28)	0.030	0.01	<i>PRKCD</i> , <i>MYL6B</i> , <i>RAMP1</i> , <i>ADORA2A</i> , <i>EDN1</i> , <i>EDN3</i> , <i>ADCY4</i> , <i>AVPR1A</i> , <i>MYH9</i> , <i>KCNMB2</i>	<i>CALM1</i> , <i>ITPR3</i> , <i>KCNMB4</i> , <i>MAP2K2</i> , <i>PLA2G4A</i> , <i>EDN1</i> , <i>ADM</i> , <i>KCNU1</i> , <i>PLA2G1B</i> , <i>MYLK2</i> , <i>MYL6</i> , <i>MYL6B</i> , <i>MYLK3</i> , <i>PLA2G10</i>
Alanine, aspartate, and glutamate metabolism	0.02 (3.15)	0.01 (0.52)	0.049	0.95	<i>AGXT2</i> , <i>ASS1</i> , <i>GLUD1</i> , <i>ALDH5A1</i> , <i>NIT2</i> , <i>FOLH1</i>	<i>RIMKL3</i> , <i>ALDH4A1</i>

^aFDR-adjusted P value. ^bNominal (unadjusted) P value. ^cCommon genes across the pathways are underlined. ^dCommon genes across the studies are in bold. ^eFold enrichment indicates the degree to which a feature is overrepresented relative to chance: >1 = enriched; 1 = no enrichment; <1 = depleted.

FDR = false discovery rate; KEGG = Kyoto Encyclopedia of Genes and Genomes; other abbreviations as in Table 1.

additional genetic variants *CPT1C*, *CPT1B*, and *CPT1A* common to the adipocytokine and fatty acid pathways enriched in patients with resilience. These genes encode for isoforms of the carnitine/choline acetyltransferase family that facilitate transport of long-chain fatty acids into mitochondria.³¹ The expression of each isoform varies by organ. *CPT1C* is highly expressed in neuronal tissue, *CPT1B* is expressed mainly in myocardial and skeletal muscle, and *CPT1A* is expressed mainly in intestinal tissue. Consistent with these pathway-level findings, querying publicly available GWAS resources (eg, the GAS catalog: <https://www.ebi.ac.uk/gwas/genes>) identified reported associations at *CPT1A* and rs597539 with lower coronary artery calcified plaque in individuals with type 2 diabetes,³² and rs2269382 with lower circulating fibrinogen,³³ whereas no cardiovascular associations were observed for *CPT1C*. These signals are directionally concordant with the observed enrichment of fatty acid metabolism/degradation and adipocytokine signaling, supporting a protective role for variation in *CPT1A*/*CPT1B* in resilience to CAD. Future work to measure plasma fatty acids among resilient patients may further elucidate the functional role of this pathway in resilience.

Because the comparison of resilient individuals to those classified as susceptible was exploratory, we regard the additional pathways as hypothesis-generating; several lack clear biological plausibility

and may represent false positives. Notably, most pathways were consistent across analyses (7 of 10), suggesting the core metabolic and vascular signals are robust, whereas the remaining pathways warrant independent validation.

Our approach to resilience to CAD risk factors improves on prior attempts to define this clinical phenotype. While there is no universally accepted definition of resilience to CAD, prior studies have characterized patients as resilient if they had a lower-than-expected burden of atherosclerosis or its complications (eg, myocardial infarction) despite elevated clinical risk.^{7,8} While important, this approach may be limited by a reliance on a single clinical risk factor, such as age or elevated LDL cholesterol.⁸ In contrast, other studies of resilience focused on patients with extreme manifestations of CAD (eg, presentation at early onset) have yielded novel risk factors and preventive therapies (for example, statins, PCSK9, or Lp(a) inhibitors), often because of an overrepresentation of a dominant risk factor. In this analysis, we advance on these approaches through the use of a highly sensitive approach to detect CAD—central core lab-based detection of coronary artery atherosclerosis using CCTA—to identify patients with and without coronary plaque from the multicenter PROMISE clinical trial. We complement the use of these clinical trial data in a relatively low-risk population with a real-world cohort of patients at higher risk undergoing

clinical angiography from the CATHGEN cohort in order to maximize the generalizability of our findings.

There are several important strengths to our approach to resilience. First, the imaging-based definition of resilience using core-lab confirmed interpretations of CCTA data overcomes event-based definitions that may misclassify patients without events as resilient. Second, we used a wide-ranging and comprehensive set of clinical risk factors and added a PRS to improve identification of high risk. This approach considered the influence of most known clinical risk modifiers, and not just the handful included in conventional risk scores. With this comprehensive approach, we identified dysregulated molecular pathways (fatty acid metabolism and degradation, vascular smooth muscle contraction, adipocytokine signaling) associated with resilience in PROMISE. Third, these findings were confirmed in CATHGEN using invasive angiography, in a population with a much higher clinical risk profile. The reproducibility of these molecular characterizations across independent cohorts along the spectrum of risk suggests that we have identified a novel, potentially protective molecular subtype of resilience.

TRANSLATIONAL SIGNIFICANCE

Potential significance of these findings relate to enhancing the precision of cardiovascular risk stratification. By validating our results in larger, multi-ethnic cohorts—integrating coronary imaging, genome-wide data, and adjudicated cardiovascular outcomes—we can establish a robust foundation for translational applications, including more precise estimates of cardiovascular risk to tailor therapies. Moreover, while our current focus is on resilience, a complementary lens examining individuals with unexpectedly high CAD burden despite low clinical and polygenic risk—may identify unexpectedly high-risk patients for whom more intense pharmacotherapy is warranted. This strategy may be especially relevant for understanding “SMuRF-less” patients, who experience cardiovascular events in the absence of Standard Modifiable Risk Factors.³⁴

STUDY LIMITATIONS. Despite these strengths, our genetic association analyses were limited by statistical power given the small number of resilient patients across cohorts. A post hoc power analysis conducted in PROMISE (resilient cases and oCAD controls) indicated that we had adequate power

primarily to detect common variants with large effects, whereas single-variant and gene-level associations were beyond the study’s detection threshold. To preserve a stringent resilience phenotype and minimize heterogeneity, we did not expand the sample by including participants with lower clinical or genetic risk, accepting reduced power as a trade-off for phenotypic specificity. Future work leveraging larger cohorts, meta-analyses, or other aggregation strategies will be needed to identify more modest associations. As an alternative approach, however, we compared resilience against a more extreme comparator, that is, patients susceptible to oCAD (ie, at low clinical or genetic risk but with oCAD). With this more extreme comparison, 3 additional potential pathways were identified. Our analysis focused on genetic resilience to oCAD ($\geq 70\%$ stenosis on CCTA) in PROMISE and validated using a similar cut points by angiography in CATHGEN. We could not assess genetic resilience in nonobstructive CAD because genotyping/PGS were unavailable for all CCTA participants and plaque characterization was limited.¹⁴ We acknowledge the burden and risk of nonobstructive CAD^{15,35} and note that future work should incorporate nonobstructive disease as well as plaque phenotype and high-risk features that may represent distinct risk profiles for genetic association studies of resilience. Other limitations of this analysis include the use of a genetic-based approach, which may miss dynamic changes that integrate comorbidities and lifestyle factors. For example, alternative approaches such as metabolomics or proteomic profiling of resilience may identify additional metabolites, proteins, and pathways associated with resilience expressed in peripheral blood. Last, our approach a priori focused on molecular determinants of resilience, whereas there are likely other equally important, nonmolecular factors, including social determinants of health,¹⁰ that were outside the scope of the current study. Future work on resilience to CAD should incorporate these additional factors for a more comprehensive understanding of resilience or susceptibility to CAD risk factors.

CONCLUSIONS

Our proof-of-concept approach rigorously characterized patients with high clinical and genetic risk yet absence of disease in order to define a robust, testable phenotype of resilience. We leveraged core lab-confirmed CAD phenotypes in low (PROMISE) and

high (CATHGEN) clinical risk cohorts, wide-ranging clinical risk modifiers, and PRSs from study biorepositories to identify putative molecular pathways underlying resilience. This modular approach may be scalable to larger, multiethnic data sets and could accelerate identification of causal pathways and therapeutic targets. We found that resilience to CAD was characterized by alterations in common metabolic and vascular pathways and shared across 2 independent cohorts, including adipocytokine signaling, fatty acid metabolism/degradation, and vascular smooth muscle contraction. Future work can expand on our approach to better understand resilience and extend these methods to characterize susceptibility to oCAD and cardiovascular risk factor and evaluate the implications for novel diagnostic and therapeutic approaches based on these pathways.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: Understanding resilience to CAD—defined as the absence of coronary plaque despite high clinical and polygenic risk—advances medical knowledge by linking imaging-defined phenotypes with protective metabolic and vascular pathways (adipocytokine signaling, fatty acid metabolism/degradation, vascular smooth muscle contraction). These insights refine risk assessment and may advance biomarker discovery and therapeutic targeting based on genomics and cardiac imaging.

TRANSLATIONAL OUTLOOK: This proof-of-concept study links imaging-defined resilience to CAD to protective metabolic and vascular pathways. Future studies may include larger, multiethnic cohorts linking imaging and genetic assessments, incorporation of phenotypes of nonobstructive plaque, and integration of multi-omics to refine risk prediction and prioritize therapeutic development.

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APPENDIX For supplemental tables and figures, please see the online version of this paper.